27/7/26

EXP: 9.

Implementation of Subnetting in Cisco Packet tracer. Simulator.

Classless Ip subnetting is a technique that allows for more efficient use of Ip address by allowing subnet mask. that are not just default mask for each Ip class.

Creating a network topology

The first step in classless Ip to create network topology in packet tracer. To create a network topology, select "New" button. then select "Network", "Generic", this will create an blank topology

Adding the device

After creating topology, we can add device. we will add switch, router. PC.

Choose the device from the bottom left corner. Connect the devices by dragging a cable from one device's port to another.

Subnetting:

To subnet the network address 192.168.1.0/24

to provide atleast 5 address for end device. the switch router, we can use a subnet mask.

the Ip addressing for the network is.

* Router R1:

gigabit ethernet 0/0 : 192.168.1.1

gigabit ethernet 0/1 : 192.168.2.1

Switch S1:

FastEthernet 0/1 : 192.168.1.0/27

PC1 : 192.168.1.11

PC2 : 192.168.1.12

PC3 : 192.168.1.13

:

PC5 : 192.168.1.15.

Similarly for the router R0 and the PC's Ip's should be assigned.

Configuring the device:

After adding and connecting the device, we can start configuring by typing commands in CLI

#enable

configure terminal

interface FastEthernet 0/0

ip address {Ip address} {Subnet mask}

no shutdown

Exit

interface fastEthernet 0/1

ip address {Ip address} {subnet mask}

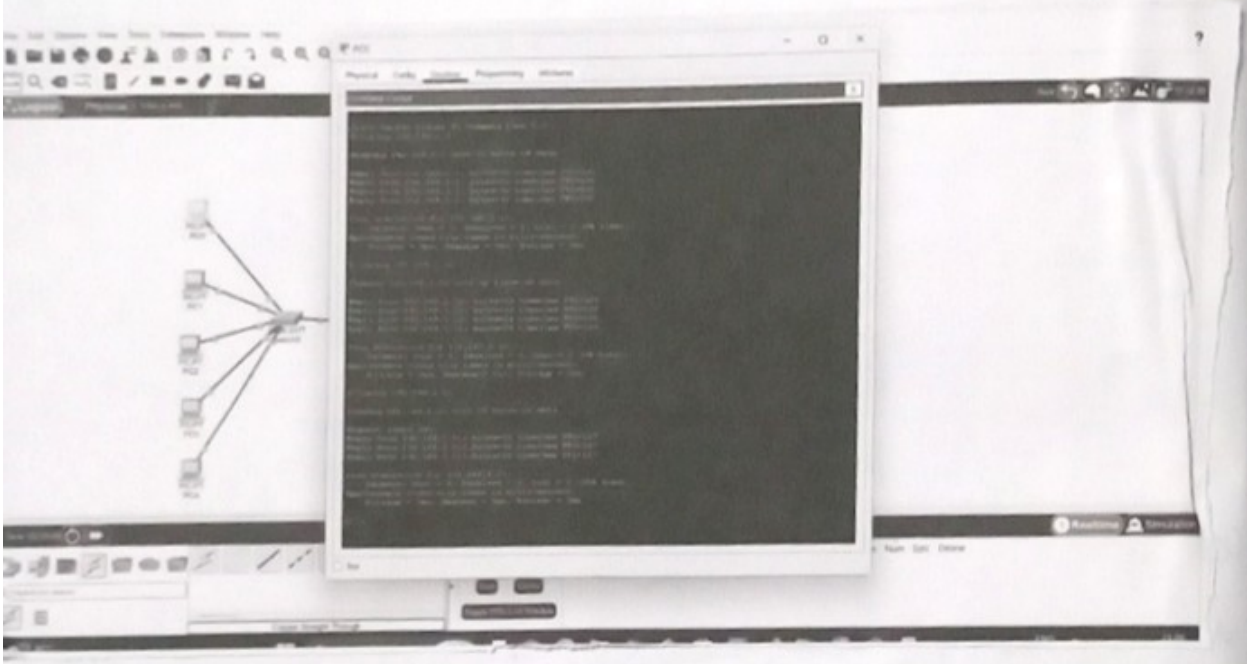
no shutdown

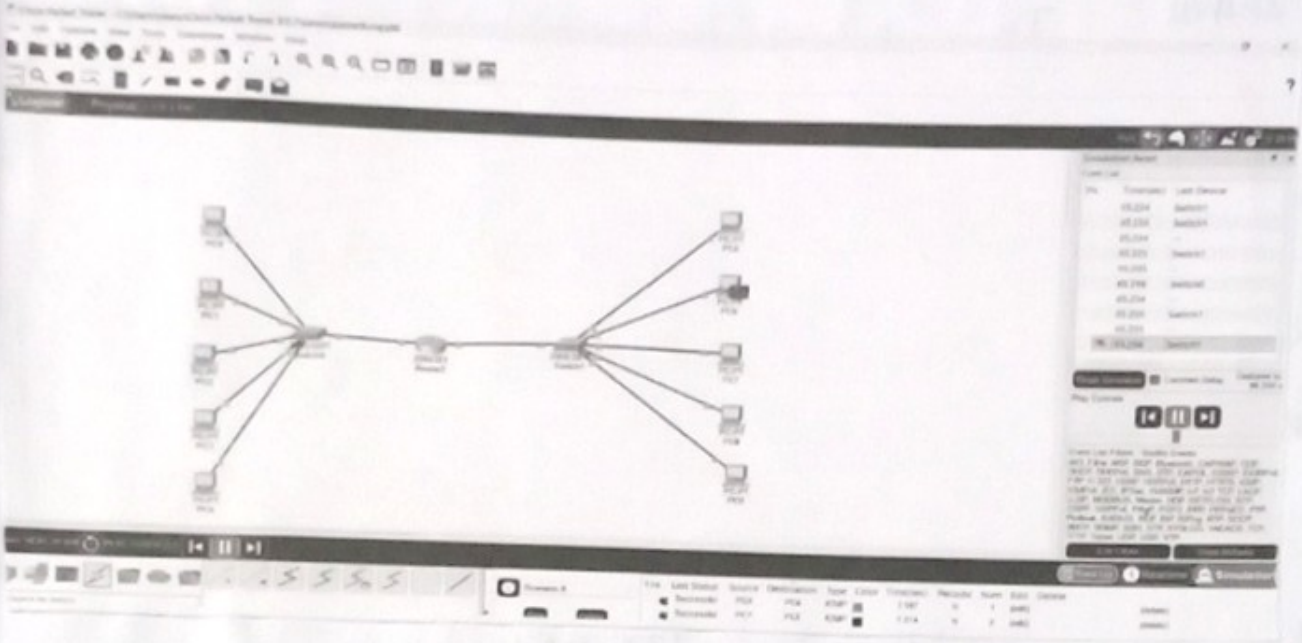
exit

Testing the network

open a command prompt on each pc

try to ping other pc. If ping is successful, then network is functioning properly.





Result:

Therefore Implementation of subnetting using cisco packet tracer executed successfully.